

量子與統計. [第二次作業]

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廖淑鋒. 微引. 集成電路基礎 2019.10.996

1.

~~波函數及其統計解釋~~

~~波函數~~

波函數與 Schrödinger 方程

-(1) 波函數及其統計解釋: ~~波函數~~

(i) 波粒二象性分析: 單個電子便有波動性

(一) 波動性: "可疊加性", "干涉", "衍射", "偏振" 等現象

(二) 粒子性: "顆粒性" 或 "整體性"

(ii) 解釋: 波函數在空間中某一點強度 (振幅絕對值平方) 和在該點找到粒子概率密度成正比例

-(2) 態疊加原理: 若 ψ_1 和 ψ_2 都是體系的可能狀態, 那麼, 它們的線性疊加

$$\Psi = c_1 \psi_1 + c_2 \psi_2 \quad (\forall c_1, c_2 \in \mathbb{C}) \text{ 也是體系的可能狀態}$$

-(3) Schrödinger 方程:

(i) 性質: ~~線性~~ 線性方程、方程系數不含狀態參量

(ii) 方程:

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \Psi + U(\vec{r}) \Psi$$

(iii) 定態方程: 若 $\frac{\partial U}{\partial t} = 0$, 則設 $\Psi(\vec{r}, t) = f(t) \varphi(\vec{r})$

$$\begin{cases} i\hbar \frac{df}{dt} = E f(t) \Leftrightarrow f(t) = C e^{-i \frac{E}{\hbar} t} \\ -\frac{\hbar^2}{2m} \nabla^2 \varphi + U(\vec{r}) \varphi = E \varphi \end{cases}$$

$$\Psi = \varphi(\vec{r}) e^{-i \frac{E}{\hbar} t}$$

(iv) 概率守恆定律: $w = |\Psi|^2 = \Psi^* \Psi$

$$\frac{\partial w}{\partial t} + \left[\frac{i\hbar}{2m} \nabla \cdot (\Psi \nabla \Psi^* - \Psi^* \nabla \Psi) \right] = \frac{\partial w}{\partial t} + \nabla \cdot \vec{J} = 0$$

$$\frac{\partial}{\partial t} \int_V w dV = - \int_V \nabla \cdot \vec{J} dV = - \int_S \vec{J} \cdot d\vec{s}$$

~~引理~~

-(4) 一維問題一般分析

-(1) 共軛定理: 若 ψ 為定態 Schrödinger 方程的解, 則 ψ^* 也是.

-(2) 實解定理: 對於能量 E 的本征值, 總存在一組實解. 令 E 的所有解都是這組解的線性疊加.

-(3) 反射定理: 若 $V(x) = V(-x)$, 則 ~~若 $\psi(x)$ 是方程的解~~ $\varphi(-x)$ 也是方程的解

-(4) 宇稱: $\begin{cases} \varphi(x) = \varphi(-x) \text{ 正宇稱} \\ \varphi(x) = -\varphi(-x) \text{ 負宇稱} \end{cases}$

引理: 任何一組解都可以被展開成確守宇稱的線性疊加

(5) Wronskian 定理:

若 ψ_1 和 ψ_2 都是以 E 为本征值的定态 Schrodinger 方程的解

则 $\begin{vmatrix} \psi_1' & \psi_2' \\ \psi_1 & \psi_2 \end{vmatrix} = C$ (C 为常数
 ψ_1, ψ_2 线性无关 $\Leftrightarrow C \neq 0$)

(6) 不简并定理: 若 $V(x)$ 无奇点, 且存在束缚态, 则必为非简并.

(5) 一维无限深方阱和有限深势阱

(i) 一维无限深方阱: $\psi_n(x) = \begin{cases} \sqrt{\frac{2}{a}} \sin(\frac{n\pi}{a}x), & 0 < x < a \\ 0, & x < 0, x > a \end{cases}$
 $E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2}$

(ii) 有限深对称势阱:

- 偶宇称: $\psi(x) = \begin{cases} B \exp(-\frac{\sqrt{2m(V_0-E)}}{\hbar}|x|), & |x| > a \\ C \cos(\frac{\sqrt{2mE}}{\hbar}x), & |x| \leq a \end{cases}$
 $\tan(\frac{\sqrt{2mE}}{\hbar}a) = \sqrt{\frac{V_0}{E}-1}$

- 奇宇称:

$\psi(x) = \begin{cases} B \exp(-\frac{\sqrt{2m(V_0-E)}}{\hbar}|x|) \operatorname{sgn}(x), & |x| > a \\ D \sin(\frac{\sqrt{2mE}}{\hbar}x), & |x| \leq a \end{cases}$
 $\cot(\frac{\sqrt{2mE}}{\hbar}a) = -\sqrt{\frac{V_0}{E}-1}$

隧穿效应:

$\psi(x) = \begin{cases} \exp(ikx) + R \exp(-ikx), & x < 0 \\ A \exp(kx) + B \exp(-kx), & 0 < x < a \\ S \exp(ikx), & x > a \end{cases}$

$k = \frac{\sqrt{2mE}}{\hbar}, \quad \kappa = \frac{\sqrt{2m(V_0-E)}}{\hbar}$

入射概率流密度: $j_{\text{in}} = \frac{\hbar k}{m}$

反射: $j_{\text{r}} = |R|^2 \frac{\hbar k}{m}$ 透射: $j_{\text{t}} = |S|^2 \frac{\hbar k}{m}$

$T = |S|^2 = \left[1 + \frac{1}{\frac{4E}{V_0} (1 - \frac{E}{V_0})} \sinh^2 \kappa a \right]^{-1}$ $|R|^2 = \frac{(\kappa^2 + k^2) \sinh^2 \kappa a}{(\kappa^2 + k^2) \sinh^2 \kappa a + 4k^2 \kappa^2}$

线性谐振子:

$E_n = \hbar \omega (n + \frac{1}{2}), \quad n = 0, 1, 2, \dots$

$\psi_n(x) = \left(\frac{\alpha}{\pi^{1/2} 2^n n!} \right)^{1/2} e^{-\frac{1}{2} \alpha^2 x^2} H_n(\alpha x), \quad \alpha = \sqrt{\frac{m\omega}{\hbar}}$

$H_n(x) = \frac{(-1)^n}{2^n n!} \frac{d^n}{dx^n} e^{-x^2}$

量子統計

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2. (2.1) 在定態中 $\psi(r, t)$

在定態中，設各能值是 $E_1, E_2, \dots, E_n, \dots$ ，對應的波函數是 $\psi_1, \psi_2, \dots, \psi_n, \dots$

$$\psi_n(r, t) = \psi_n(r) e^{-i \frac{E_n}{\hbar} t}$$

因此，在此定態中，其任一概率波函數 $\psi = \sum_n c_n \psi_n = \sum_n c_n \psi_n(r) e^{-i \frac{E_n}{\hbar} t}$

$$\text{概率流密度 } \vec{j} = \frac{i\hbar}{2m} (\psi \nabla \psi^* - \psi^* \nabla \psi)$$

=

2. (2.1) 對於任一態，都有一個能量本征值 E

其波函數 $\psi(r, t) = \psi(r) e^{-i \frac{E}{\hbar} t}$

$$\text{概率流密度 } \vec{j} = \frac{i\hbar}{2m} (\psi \nabla \psi^* - \psi^* \nabla \psi)$$

$$= \frac{i\hbar}{2m} [\psi(r) e^{-i \frac{E}{\hbar} t} \cdot e^{i \frac{E}{\hbar} t} \nabla \psi^*(r) - \psi^*(r) e^{i \frac{E}{\hbar} t} \cdot e^{-i \frac{E}{\hbar} t} \nabla \psi(r)]$$

$$= \frac{i\hbar}{2m} (\psi \nabla \psi^* - \psi^* \nabla \psi), \text{ 與時間 } t \text{ 無關}$$

3. (2.2) 利用上題結論

$$(1) \vec{j}_1 = \frac{i\hbar}{2m} (\psi_1 \nabla \psi_1^* - \psi_1^* \nabla \psi_1)$$

$$= \frac{i\hbar}{2m} (\psi_1 \frac{d}{dr} \psi_1^* - \psi_1^* \frac{d}{dr} \psi_1) \vec{e}_r$$

$$= \frac{i\hbar}{2m} \left[\frac{1}{r} e^{ikr} \cdot \left(-\frac{1}{r^2} e^{-ikr} + \frac{-ik}{r} e^{-ikr} \right) - \frac{1}{r} e^{-ikr} \cdot \left(-\frac{1}{r^2} e^{ikr} + \frac{ik}{r} e^{ikr} \right) \right] \vec{e}_r$$

$$= \frac{i\hbar}{2m} \left[\frac{1}{r} \left(\frac{ik}{r} + \frac{1}{r^2} \right) - \frac{1}{r} \left(\frac{-ik}{r} - \frac{1}{r^2} \right) \right] \vec{e}_r$$

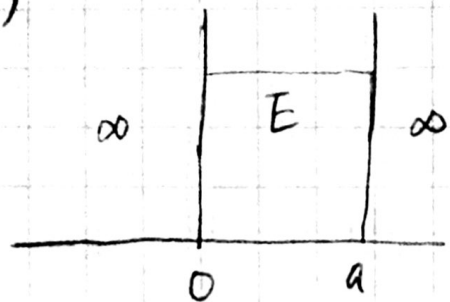
$$= \frac{i\hbar}{2m} \left(-\frac{ik}{r^2} - \frac{1}{r^3} - \frac{ik}{r^2} + \frac{1}{r^3} \right) \vec{e}_r = -\frac{2i^2 \hbar k}{2m \cdot r^2} = \frac{\hbar k}{2m} \vec{e}_r$$

$$= \frac{\hbar k}{2m} \vec{e}_r$$

$$(2) \vec{j}_2 = \frac{i\hbar}{2m} (\psi_2 \nabla \psi_2^* - \psi_2^* \nabla \psi_2)$$

$$= \frac{i\hbar}{2m} (\psi_1^* \nabla \psi_1 - \psi_1 \nabla \psi_1^*) = -\vec{j}_1 = -\frac{\hbar k}{2m} \vec{e}_r$$

4 (2.3)



$$\begin{cases} -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) = E \psi(x) \Rightarrow \ddot{\psi}(x) + \frac{2mE}{\hbar^2} \psi(x) = 0, & 0 \leq x \leq a \\ \psi(x) = 0, & (x < 0 \text{ 或 } x > a) \end{cases}$$

$$\begin{cases} \psi(x) = 0, & x < 0 \text{ 或 } x > a \\ \psi(x) = A \sin\left(\sqrt{\frac{2mE}{\hbar^2}} x + \delta\right), & 0 \leq x \leq a \end{cases}$$

結合波函數的連續性，得 $\psi(0) = \psi(a) = 0$

$$\Rightarrow \delta = 0 \text{ 且 } \frac{\sqrt{2mE}}{\hbar} = \frac{n\pi}{a} \quad (n \in \mathbb{Z}) \Leftrightarrow \delta = 0 \text{ 且 } E_n = \frac{n^2 \hbar^2 \pi^2}{2ma^2}$$

再考慮波函數的歸一性，得 $\frac{A^2}{2} a = 1 \Rightarrow |A| = \sqrt{\frac{2}{a}}$ ，取 $A = \sqrt{\frac{2}{a}}$ 。

所以粒子的能級是 $E_n = \frac{n^2 \hbar^2 \pi^2}{2ma^2} \quad (n=1, 2, \dots)$

對應波函數是 $\Psi_n(x, t) = \psi_n(x) \exp(-i \cdot \frac{E_n}{\hbar} t)$

$$= \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a} x\right) e^{-i \cdot \frac{n^2 \hbar^2 \pi^2}{2ma^2} t}$$

5 (2.5)

$$\psi_n(x) = \sqrt{\frac{\alpha}{\pi^{1/2} 2^n n!}} e^{-\frac{1}{2}\alpha^2 x^2} H_n(\alpha x), \quad n=1, \quad \alpha = \sqrt{\frac{m\omega}{\hbar}}$$

$$\xi = \alpha x, \quad H_1(\xi) = 2\xi$$

當 $x > 0$ 時 若 $\frac{d}{dx} \psi_1(x) = 0$, 則 $\frac{d}{d\xi} [\exp(-\frac{\xi^2}{2}) \cdot 2\xi] = 0$

$$\exp(-\frac{\xi^2}{2}) \cdot (-\xi) \cdot 2\xi + \exp(-\frac{\xi^2}{2}) \cdot 2 = 0$$

$$\Rightarrow \xi^2 = 1 \Rightarrow \xi = 1 = \alpha x \Rightarrow x_m = \frac{1}{\alpha} = \sqrt{\frac{\hbar}{m\omega}}$$

當 $x \leq 0$ 時 $\psi_1(x) = -\psi_1(-x)$

$$\frac{d}{dx} \psi_1(x) = -\frac{d}{dx} \psi_1(-x) = -\frac{d}{d(-x)} \psi_1(-x) = 0$$

$$\Rightarrow -x = \frac{1}{\alpha} \Rightarrow x_m = -\frac{1}{\alpha} = -\sqrt{\frac{\hbar}{m\omega}}$$

$\Rightarrow$ 第一激發態概率最大位置為 $x = \pm \sqrt{\frac{\hbar}{m\omega}}$

6 (2.6)

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) + U(x) \psi(x) = E \psi(x)$$

$$\Leftrightarrow -\frac{\hbar^2}{2m} \frac{d^2}{d(-x)^2} \psi(-x) + U(-x) \psi(-x) = E \psi(-x)$$

$$\text{因為 } \frac{d^2}{d(-x)^2} \psi(-x) = \frac{d}{d(-x)} \left[-\frac{d}{dx} \psi(-x) \right] = \frac{d}{dx} \left(\frac{d}{dx} \psi(-x) \right) = \frac{d^2}{dx^2} \psi(-x)$$

$$U(-x) = U(x)$$

所以 $-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(-x) + U(x) \psi(-x) = E \psi(-x)$, 即 $\psi(-x)$ 也是一個能量為 E 的波函數

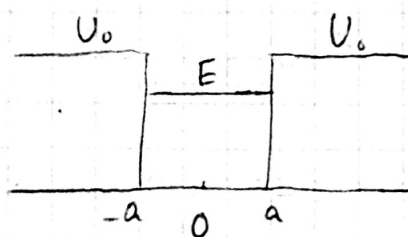
由於 $\psi(x)$ 和 $\psi(-x)$ 描述同一個勢場的波函數

所以必存在一個常數 c , 使得 $\psi(-x) = c\psi(x)$

$$\psi(x) = c\psi(-x) = c^2\psi(x) \Rightarrow c = \pm 1$$

$$\begin{cases} \psi(-x) = -\psi(x) & \text{奇宇稱} \\ \psi(-x) = \psi(x) & \text{偶宇稱} \end{cases}$$

即有確定的宇稱。



阱外: $-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) = (E - U_0) \psi(x) \Rightarrow \psi(x) = A \exp\left(\frac{\sqrt{2m(V_0 - E)}}{\hbar} x\right) + B \exp\left(-\frac{\sqrt{2m(V_0 - E)}}{\hbar} x\right)$

阱内: $-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) = E \psi(x) \Rightarrow \psi(x) = C \cos\left(\frac{\sqrt{2mE}}{\hbar} x\right) + D \sin\left(\frac{\sqrt{2mE}}{\hbar} x\right)$

当 $x \rightarrow +\infty$ 时, $\psi(x) = 0$, 所以 $A = 0$ ($x > a$), $\psi(x) = B \exp\left(-\frac{\sqrt{2m(V_0 - E)}}{\hbar} x\right)$, $x \geq a$

同理可得 $x < -a$ 时, $B = 0$, $\psi(x) = A \exp\left(\frac{\sqrt{2m(V_0 - E)}}{\hbar} x\right)$

因为 $U(x) = U(-x)$, 所以 $\psi(x)$ 具有确定的宇称。

(1) 偶宇称: $A = B, D = 0$

$$\psi(x) = \begin{cases} B \exp\left(-\frac{\sqrt{2m(V_0 - E)}}{\hbar} |x|\right), & |x| > a \\ C \cos\left(\frac{\sqrt{2mE}}{\hbar} x\right), & |x| \leq a \end{cases}$$

利用 ψ 的连续性和 ψ' 的连续性, 得

$$\begin{cases} B \exp\left(-\frac{\sqrt{2m(V_0 - E)}}{\hbar} a\right) = C \cos\left(\frac{\sqrt{2mE}}{\hbar} a\right) \\ -\frac{\sqrt{2m(V_0 - E)}}{\hbar} B \exp\left(-\frac{\sqrt{2m(V_0 - E)}}{\hbar} a\right) = -\frac{\sqrt{2mE}}{\hbar} C \sin\left(\frac{\sqrt{2mE}}{\hbar} a\right) \end{cases}$$

$$\Rightarrow \frac{\sqrt{2m(V_0 - E)}}{\hbar} = \frac{\sqrt{2mE}}{\hbar} \tan\left(\frac{\sqrt{2mE}}{\hbar} a\right) \Rightarrow \tan\left(\frac{\sqrt{2mE}}{\hbar} a\right) = \sqrt{\frac{V_0}{E} - 1}$$

(2) 奇宇称: $A = -B, C = 0$

$$\psi(x) = \begin{cases} B \exp\left(-\frac{\sqrt{2m(V_0 - E)}}{\hbar} x\right) \operatorname{sgn}(x), & |x| > a \\ D \sin\left(\frac{\sqrt{2mE}}{\hbar} x\right), & |x| \leq a \end{cases}$$

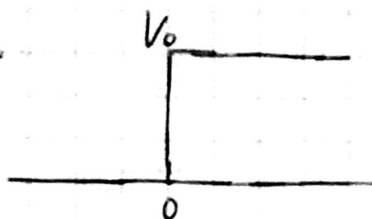
同(1)一样, $B \exp\left(-\frac{\sqrt{2m(V_0 - E)}}{\hbar} a\right) = D \sin\left(\frac{\sqrt{2mE}}{\hbar} a\right)$

$$-\frac{\sqrt{2m(V_0 - E)}}{\hbar} B \exp\left(-\frac{\sqrt{2m(V_0 - E)}}{\hbar} a\right) = \frac{\sqrt{2mE}}{\hbar} D \cos\left(\frac{\sqrt{2mE}}{\hbar} a\right)$$

$$\Rightarrow -\frac{\sqrt{2m(V_0 - E)}}{\hbar} = \frac{\sqrt{2mE}}{\hbar} \cot\left(\frac{\sqrt{2mE}}{\hbar} a\right)$$

$$\Rightarrow \cot\left(\frac{\sqrt{2mE}}{\hbar} a\right) = -\sqrt{\frac{V_0}{E} - 1}$$

8.



$$(1) \begin{cases} -\frac{\hbar^2}{2\mu} \frac{d^2}{dx^2} \psi(x) = E \psi(x), & x < 0 \\ -\frac{\hbar^2}{2\mu} \frac{d^2}{dx^2} \psi(x) + V_0 \psi(x) = E \psi(x), & x > 0 \end{cases}$$

$$\Leftrightarrow \begin{cases} \ddot{\psi}(x) + \frac{2\mu E}{\hbar^2} \psi(x) = 0, & x < 0 \\ \ddot{\psi}(x) + \frac{2\mu(E-V_0)}{\hbar^2} \psi(x) = 0, & x > 0 \end{cases}$$

$$\Leftrightarrow \psi(x) = \begin{cases} A e^{ik_1 x} + A' e^{-ik_1 x}, & k_1 = \frac{\sqrt{2\mu E}}{\hbar}, & x < 0 \\ B e^{ik_2 x}, & k_2 = \frac{\sqrt{2\mu(E-V_0)}}{\hbar}, & x > 0 \end{cases}$$

基於 ψ 和 $\dot{\psi}$ 的連續性 (在 $x=0$ 處).

$$\begin{cases} A + A' = B \\ k_1 A - k_1 A' = k_2 B \end{cases} \Leftrightarrow \begin{cases} A' = \frac{k_1 - k_2}{k_1 + k_2} A \\ B = \frac{2k_1}{k_1 + k_2} A \end{cases}$$

反射

$$\text{概率 } T = \left| \frac{A'}{A} \right|^2 = \left(\frac{k_1 - k_2}{k_1 + k_2} \right)^2 = \left(\frac{\sqrt{E-V_0} - \sqrt{E}}{\sqrt{E-V_0} + \sqrt{E}} \right)^2$$

$$\text{透射概率 } R = \left| \frac{B}{A} \right|^2 = \left(\frac{2k_1}{k_1 + k_2} \right)^2 = \left(\frac{\sqrt{E}}{\sqrt{E-V_0} + \sqrt{E}} \right)^2$$

透射

$$(2) \begin{cases} \ddot{\psi}(x) - \frac{2\mu(V_0-E)}{\hbar^2} \psi(x) = 0, & x > 0 \\ \ddot{\psi}(x) + \frac{2\mu E}{\hbar^2} \psi(x) = 0, & x < 0 \end{cases}$$

$$\Leftrightarrow \psi(x) = \begin{cases} C e^{-k_3 x}, & k_3 = \frac{\sqrt{2\mu(V_0-E)}}{\hbar}, & x > 0 \\ A e^{ik_1 x} + A' e^{-ik_1 x}, & k_1 = \frac{\sqrt{2\mu E}}{\hbar}, & x < 0 \end{cases}$$

基於 ψ 和 $\dot{\psi}$ 的連續性 ($x=0$ 處)

$$\begin{cases} A + A' = C \\ ik_1(A - A') = -k_3 C \end{cases} \Leftrightarrow \begin{cases} A' = \frac{ik_1 + k_3}{ik_1 - k_3} A \\ C = \frac{2ik_1}{ik_1 + k_3} A \end{cases}$$

$$J_R = -\frac{\hbar k_1}{\mu} |A|^2 = \frac{\hbar k_1}{\mu} \cdot \frac{k_3^2 + k_1^2}{k_3^2 + k_1^2} |A|^2 = \frac{\hbar k_1}{\mu} |A|^2$$

$$J_T = \frac{\hbar k_3}{\mu} |C|^2 e^{-2k_3 x} (k_3) - |C|^2 e^{-2k_3 x} (-k_3) = 0$$

反射概率為 1, 透射概率為 0

$$9. (1) \hat{H} = \frac{\hbar^2}{2m} \nabla^2 + V = -\frac{\hbar^2}{2m} \nabla^2$$

$$\hat{H} = -\frac{\hbar^2}{2m} \nabla^2 + V = \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V_x \right) + \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial y^2} + V_y \right) \\ = \hat{H}_x + \hat{H}_y$$

$$\text{其中 } V_x = \frac{1}{2} m \omega^2 x^2, \quad V_y = \frac{1}{2} m \omega^2 y^2$$

利用分离变量法, 设 $\psi(x, y) = \psi_I(x) \psi_{II}(y) = \psi_x \psi_y$

$$\hat{H} \psi = (\hat{H}_x + \hat{H}_y) \psi_I(x) \psi_{II}(y) = (E_I \psi_I(x)) \psi_{II}(y) + \psi_I(x) (E_{II} \psi_{II}(y)) \\ = (E_I + E_{II}) \psi_I(x) \psi_{II}(y) \\ = (E_I + E_{II}) \psi \\ = E \psi$$

$$\Rightarrow E_{I, II} = E_I + E_{II}, \text{ 简并度是 } N+1$$

波函数为 $\psi_I(x) \psi_{II}(y)$

$$(2) \frac{1}{2} V(x, y) = \frac{1}{2} m \omega^2 (x^2 + xy + y^2) = \frac{1}{2} m \omega^2 \left(\frac{3}{2} u^2 + \frac{1}{2} v^2 \right)$$

$$\frac{\partial}{\partial x} = \frac{\partial}{\partial u} \cdot \frac{1}{\sqrt{2}} + \frac{\partial}{\partial v} \cdot \frac{1}{\sqrt{2}}$$

$$\frac{\partial}{\partial y} = \frac{\partial}{\partial u} \cdot \frac{1}{\sqrt{2}} - \frac{\partial}{\partial v} \cdot \frac{1}{\sqrt{2}}$$

$$\begin{cases} \frac{\partial^2}{\partial x^2} = \frac{1}{2} \frac{\partial^2}{\partial u^2} + \frac{1}{2} \frac{\partial^2}{\partial v^2} + \frac{1}{2} \frac{\partial^2}{\partial u \partial v} + \frac{1}{2} \frac{\partial^2}{\partial v \partial u} \\ \frac{\partial^2}{\partial y^2} = \frac{1}{2} \frac{\partial^2}{\partial u^2} + \frac{1}{2} \frac{\partial^2}{\partial v^2} - \frac{1}{2} \frac{\partial^2}{\partial u \partial v} - \frac{1}{2} \frac{\partial^2}{\partial v \partial u} \end{cases} \Rightarrow \nabla_{xy}^2 = \nabla_{uv}^2$$

$$\hat{H} = -\frac{\hbar^2}{2m} \nabla_{xy}^2 + \frac{1}{2} V_{xy} = -\frac{\hbar^2}{2m} \nabla_{uv}^2 + \frac{3}{2} V_u + \frac{1}{2} V_v = \hat{H}_u(\sqrt{\frac{3}{2}} \omega) + \hat{H}_v(\sqrt{\frac{1}{2}} \omega)$$

波函数 ψ

$$\text{所以设 } \psi(u, v) = \psi_I(\sqrt{\frac{3}{2}} \omega, u) \psi_{II}(\sqrt{\frac{1}{2}} \omega, v)$$

$$\hat{H} \psi = [E_I(\sqrt{\frac{3}{2}} \omega) + E_{II}(\sqrt{\frac{1}{2}} \omega)] \psi = E \psi, \text{ 简并度 } N+1$$

$$\text{其中 } \hat{H}_u(\omega) = -\frac{\hbar^2}{2m} \frac{d^2}{du^2} + \frac{1}{2} m \omega^2 u^2$$

$$\psi_n(u, u) = \left(\frac{\sqrt{\frac{m\omega}{\hbar}}}{\pi^{1/2} 2^n n!} \right)^{1/2} e^{-\frac{1}{2} \frac{m\omega}{\hbar} u^2} H_n \left(\sqrt{\frac{m\omega}{\hbar}} u \right)$$

$$E_n(\omega) = \left(n + \frac{1}{2} \right) \hbar \omega$$